

Title : Project Registration & Progress Review**FF No. 180**

Department: AIDS	Academic Year: 2025-26
Semester: II	Group No. : TY-A-G14
Project Title: Optimization and Deployment of Privacy-Preserving SLMs on Resource Constrained Edge Hardware	
Project Area: Edge AI, Model Compression, and Data Privacy	

Group Members Details:

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Project approved / Not approved

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Prof.(Dr.) Shital Dongre

Guide**Project Coordinator****Head, Dept. of AIDS**

Project Synopsis:

1. Motivation & Problem Statement: Current Generative AI relies on centralized cloud infrastructure, creating significant data privacy risks and high latency. Furthermore, the massive energy consumption of standard data centers is becoming unsustainable. There is a critical technical need for "Small-scale AI models" that provide localized, private intelligence without requiring constant cloud connectivity.

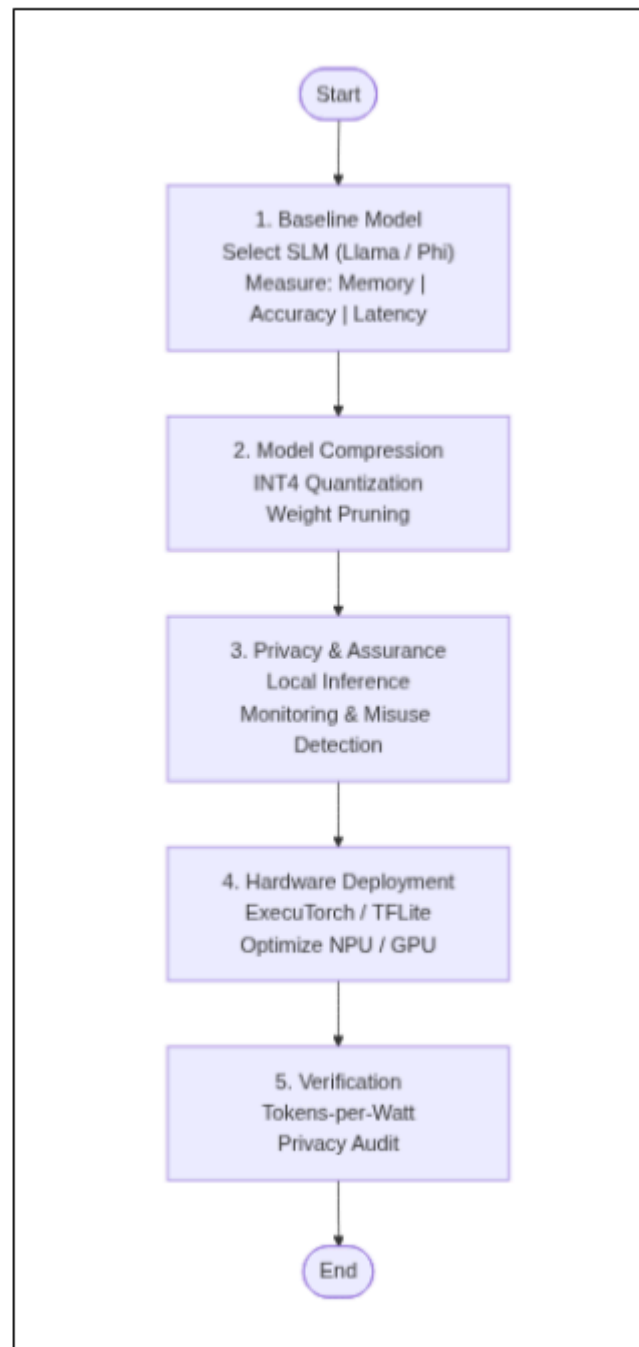
2. Objectives:

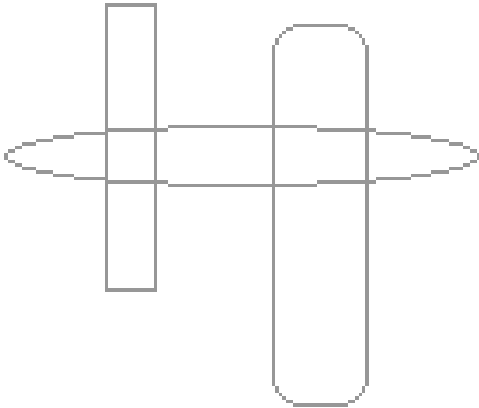
- To implement advanced **Model Compression** techniques, including **INT4 Quantization** and **Weight Pruning**, to reduce the memory footprint of a 1B–3B parameter model by at least 70%.
- To develop a **Local Inference Pipeline** that ensures 100% of user data remains on-device, fulfilling "Privacy-by-Design" principles.
- To conduct an **Energy Efficiency Analysis** by measuring "Tokens-per-Watt" and thermal dissipation on target edge hardware (e.g., Raspberry Pi 5 or Android NPU).

3. Technical Methodology:

- **AI & Optimization:** Selection of an open-weight base model (e.g., Llama 3.2 1B) followed by hardware-aware quantization to fit 4GB–8GB RAM constraints.
- **Architecture & Security:** Implementation of an **Assurance Layer** for real-time monitoring of model reliability and security.
- **Deployment & Verification:** Utilizing edge-native frameworks (e.g., ExecuTorch) to accelerate NPU performance and benchmarking results against cloud-based alternatives.

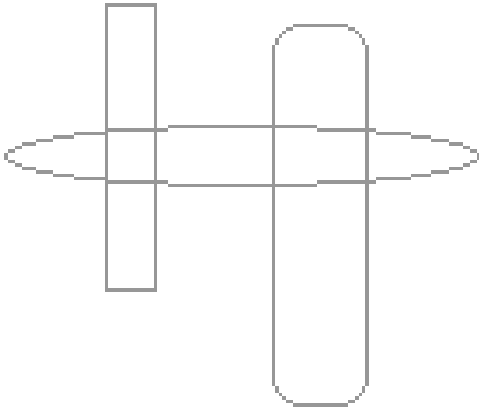
Methodology Flowchart:





Group No.			
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	Group No.:	Date:
Progress Review Report		
		

Signature of Guide:

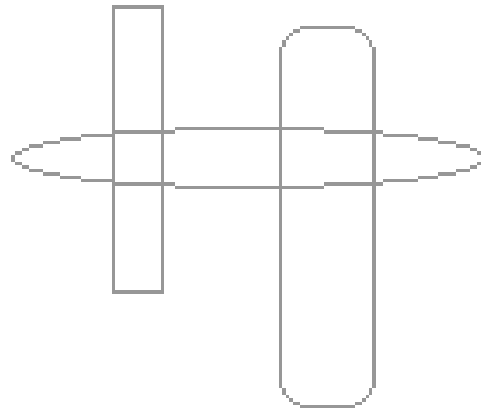
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Review No.: 2

Group No.:

Date:

Progress Review Report



Signature of Guide:

